

Object Class in java

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Date:

YOUVA
PILLAR

Object class → Parent of all classes

- java.lang (System)
- Parent → Object
- Every class in java inherits from object class

```
class Student extends Human {
```

```
}
```

class Student extends Object

```
}
```

```
class Human extends Object {
```

```
}
```

Object

→ Common behaviours (inheritance)

```
class Object  
{
```

```
class Student extends Student {  
}
```

→ Object obj = new Student();
(polymorphism)

class Object {

}

Core Methods

- toString()
- equals()
- hashCode()
- getClass()

Cloning

↳ clone()

garbage collection

↳ finalize()

Threads

- ↳ wait()
- ↳ notify()
- ↳ notifyAll()

* toString()

↳ convert anything to String

```
public String toString() {  
    {  
}
```

String representation of any object

class Object {

public String toString() {

// String Representation of Object

}

}

Student s1 = new Student();
s1.toString();

↓
className@Hexcode

class Student extends Object {

@Override

toString() {

{
}

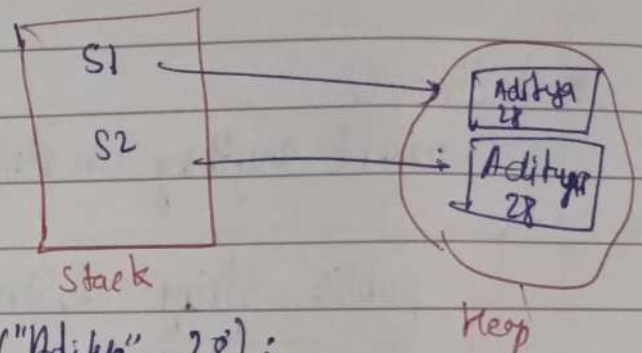
* equals

- Compare 2 objects & return true/false

```
class Object {  
    public boolean equals(Object obj) {  
        // Default  
    }  
}
```

↳ compare references

```
class Student {  
    String name;  
    int age;  
}
```



```
Student s1 = new Student("Aditya", 28);  
Student s2 = new Student("Aditya", 28);
```

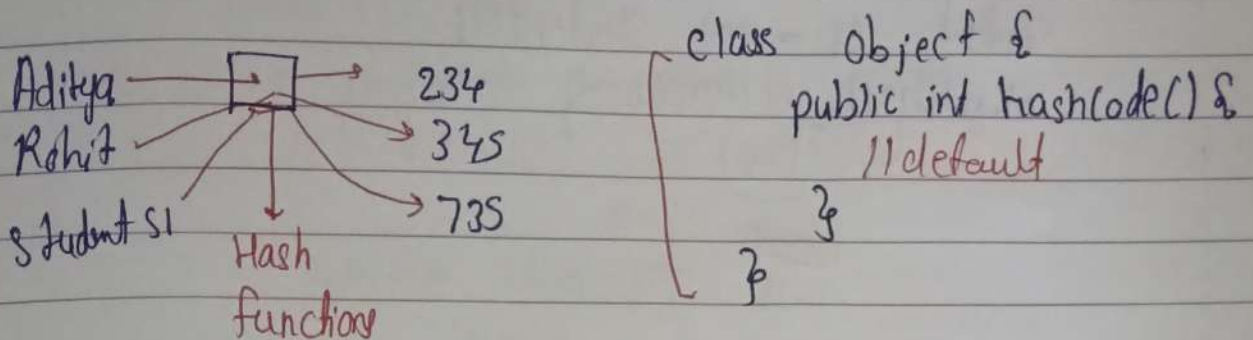
```
s1.equals(s2); // false
```

↳ because references are different

```
s1 == s2 // reference  
         compare  
false
```


* hashCode()

↳ return an integer of an Object
↳ Hexadecimal format



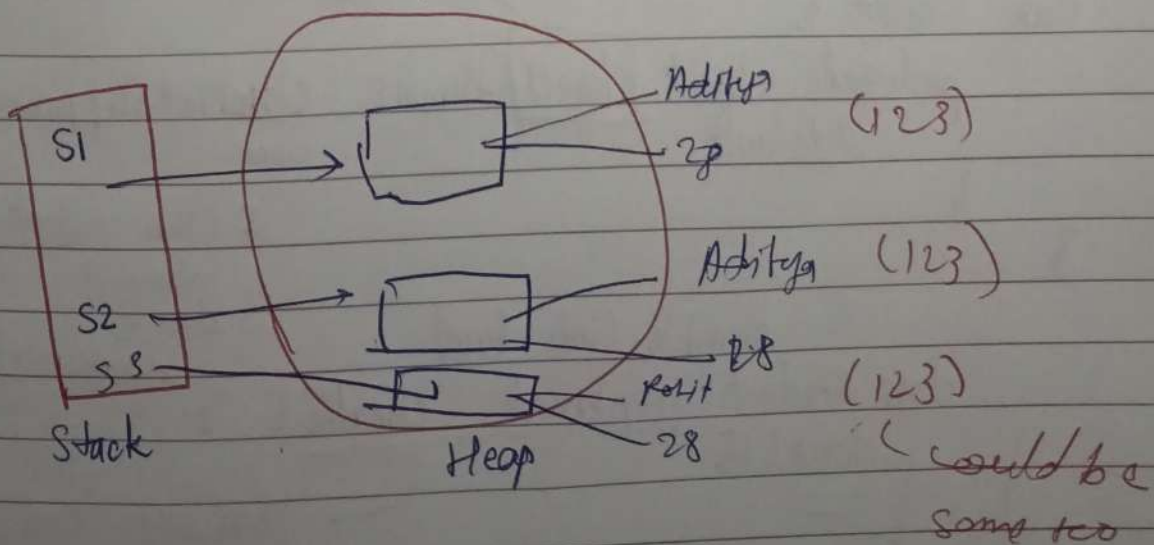
Rule:

↳ if 2 objects are equal then their hashCode must be equal.

```
Student s1 = new Student("Aditya", 28);  
Student s2 = new Student("Aditya", 28);
```

if $s1.equals(s2) \rightarrow true$
then, $s1.hashCode() == s2.hashCode() \rightarrow true$

↳ but reverse might not be true



* getClass()

↳ return runtime class of an object

```
Student s1 = new Student();
```

```
s1.getClass().getName()
```

Class
class type object

Student

```
class Object {
```

```
public final Class <?> getClass() {
```

// default

```
}
```

```
}
```

can't override this

~~if (this instanceof Cloneable) {~~

~~allow it~~

~~return~~

~~throw new CloneNotSupportedException~~

~~}~~

* clone()

↳ create copy of an Object

Shallow copy

```
class Object {
```

```
protected Object clone() throws
```

// default

```
}
```

```
}
```

~~CloneNotSupportedException~~

clone

if (this instanceof Cloneable) {

allow it

} else {

throw Exception

Cloneable (interface)

```
class Student implements Cloneable {
```

```
clone() {
```

```
==
```

```
}
```

```
}
```

then only can
override clone
method

Not every object should be Cloneable

→ Database }
→ Thread }

class Student implements Cloneable {
// override → (Deep copy) → then only you can clone
}

interface Cloneable {
// empty
}

→ marker
interface

* finalize()

→ garbage collection
→ Now deprecated
→ unpredictable }
unsafe
unreliable

* wait()

* notify()

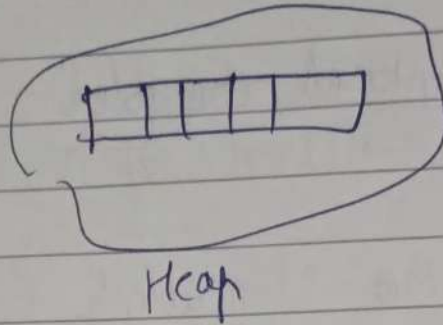
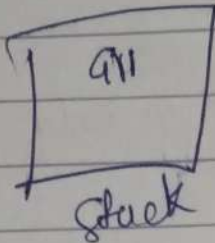
* notifyAll()

} we'll discuss in Threads

Arrays → Object
→ Non-primitive

Arrays

int[] arr = new int[5];



No object class on

int
char
bool
⋮
primitive

Wrapper classes

Integer
Boolean
Float
⋮

→
they
all
comes
under
object
class